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Project Report On

"CrowdStrike Falcon Update Failure: The Global Windows Outage of July 2024"

Submitted in partial fulfilment of the requirement for the project work for the degree of

BACHELOR OF COMPUTER APPLICATION (BCA)

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CERTIFICATE

This is to certify that the project synopsis entitled "INTELLIDOCS X – AI DOCUMENT INTELLIGENCE & VERIFICATION PLATFORM" is submitted by Ms. Sparsha S hegde (USN: U03BX24S0094) in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from Bangalore University for the academic year 2024–2027.

The project synopsis has been prepared under the guidance and supervision of the undersigned. The synopsis is submitted for academic evaluation in accordance with the applicable department and university guidelines**.**

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Abstract

CrowdStrike Global IT Outage on July 19, 2024, is considered to be one of the most remarkable cases of technological disruption in recent times. This problem happened due to the deployment of an incorrect Rapid Response Content update by CrowdStrike using their Falcon endpoint security solution to Windows machines. The incorrect update led to crashing of the machine and appearance of BSOD for those machines, which caused significant disruption to the work of various companies. According to Microsoft, about 8.5 million Windows machines were affected, and this number constituted less than one percent of all Windows machines globally. But the scale of the problem was much bigger since the machines belonged to vital companies.

The attack wasn't the usual form of cyberattack or ransomware. In fact, it occurred due to the failure of software content validation and implementation. According to Root Cause Analysis by CrowdStrike, there was an issue with input in Channel File 291 which led to out-ofbounds memory read while being processed by Falcon sensor.

This case study analyze the incident, its cause, its effect on different industries, how the incident was detected and responded to, the security measures taken, the lessons learned, and recommendations. This incident has shown that cybersecurity is not all about protecting against malicious attacks; it is also about software assurance, update control, resiliency, continuity, and recovery.

Keywords: CrowdStrike, Falcon Sensor, Windows, IT system downtime, cybersecurity, software upgrade, endpoint security, cyber resilience, software supply chain.

1. Introduction

Organizations depend heavily on IT in their day-to-day communications, financial dealings, provision of health care, logistics, governmental functions, production processes, and other business activities. In light of their growing reliance on digital infrastructure, organizations also need cybersecurity products that will protect their systems from malware, ransomware, and data breaches.

CrowdStrike is an information security firm that offers endpoint security through the Falcon product. Endpoint security software runs at a very high level of trust on a computer since it must monitor various activities, processes, files, and resources. Therefore, any malfunction of such software can lead to more than just an ordinary application problem.

The risks of using technology were evident on July 19, 2024. On that day, there was an important global technological failure that took place. This was done through an update sent by CrowdStrike called Rapid Response Content update, which is related to its Falcon platform.

It immediately developed into a worldwide issue with regard to operations. Airlines, hospitals, banks, government agencies and even companies suffered interruptions. Microsoft claimed that about 8.5 million Windows systems were infected.

A major factor in the case is the absence of a hostile attacker as the cause of the incident. It happened because of an unintentional failure during an update process. This particular situation is highly interesting in terms of cybersecurity because it shows that availability and resilience may be endangered by both attackers and mistakes in trustworthy systems.

The CrowdStrike attack thus presents a valuable chance to analyze software supply chain risk, endpoint protection, updates, testing, incident response, third party dependency, and business continuity

2. Incident Description

2.1 Background

CrowdStrike Falcon offers Endpoint Detection and Response capabilities to its customers. Rapid Response Content is one of the methods that is used by CrowdStrike for quick enhancement of the detection process. Using this method, any change related to security can be provided without installing Falcon Sensor.

Quick deployment of patches is advantageous since cyber threats constantly change. Security software vendors should have the capacity to react quickly to emerging cyber threat vectors. However, quick deployment poses some risks if the patch has not been well tested or validated.

CrowdStrike made available a Rapid Response Content update for Windows on July 19, 2024. This update was rolled out starting from around 04:09 UTC. Not long after that, crash issues started affecting Windows machines.

CrowdStrike found out the cause of the failure in the update and reversed it around 05:27 UTC. Nevertheless, computers that have processed the failed update may be in the crashed condition and would need further fixing.

2.2 Incident Timeline

Time / Date	Event
19 July 2024, 04:09 UTC	CrowdStrike began distributing the problematic content update
Shortly afterward	Affected Windows systems began experiencing crashes
During the morning	Organizations worldwide began reporting IT disruptions
05:27 UTC	CrowdStrike reverted the problematic content
19 July onward	Organizations started recovery and remediation
20 July	Microsoft published information and assistance for affected customers

This incident has proven that undoing the update on the vendor side does not always lead to restoring endpoints that have already applied and processed the

Attack Type / Vulnerability

CrowdStrike attack cannot be regarded as a typical cyber attack since there is no indication that someone had intentionally abused the flaw.

However, it may be categorized as a:

Failure of Software Update and Content Verification

The vulnerability was linked to the handling and verification of the Rapid Response Content.

CrowdStrike Root Cause Analysis reveals that the involved Falcon sensor required 20 inputs, whereas the faulty content had 21 inputs. The Content Validator did not notice this anomaly prior to release. Due to this, the Falcon sensor tried to work with the invalid data, resulting in an out-of-bounds memory read and eventually system crash.

Technical Failure Sequence

Rapid Response Content

↓

Content Validator

↓

Validation Failure

↓

Defective Content Released

↓

Falcon Sensor

↓

Invalid Memory Read

↓

System Exception

↓

Windows Blue Screen

↓

Computer Becomes Unavailable

This situation shows the need for a software application running with elevated system privileges to have effective validation and error handling.

An application error in normal circumstances could lead to one application shutting down. However, an error in an endpoint security module could have the effect of destabilizing the whole operating system.

Propagation of CrowdStrike Software Update Failure

CROWDSTRIKE

RAPID RESPONSE

CONTENT

|

▼

CONTENT VALIDATOR

|

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Validation did not detect invalid input

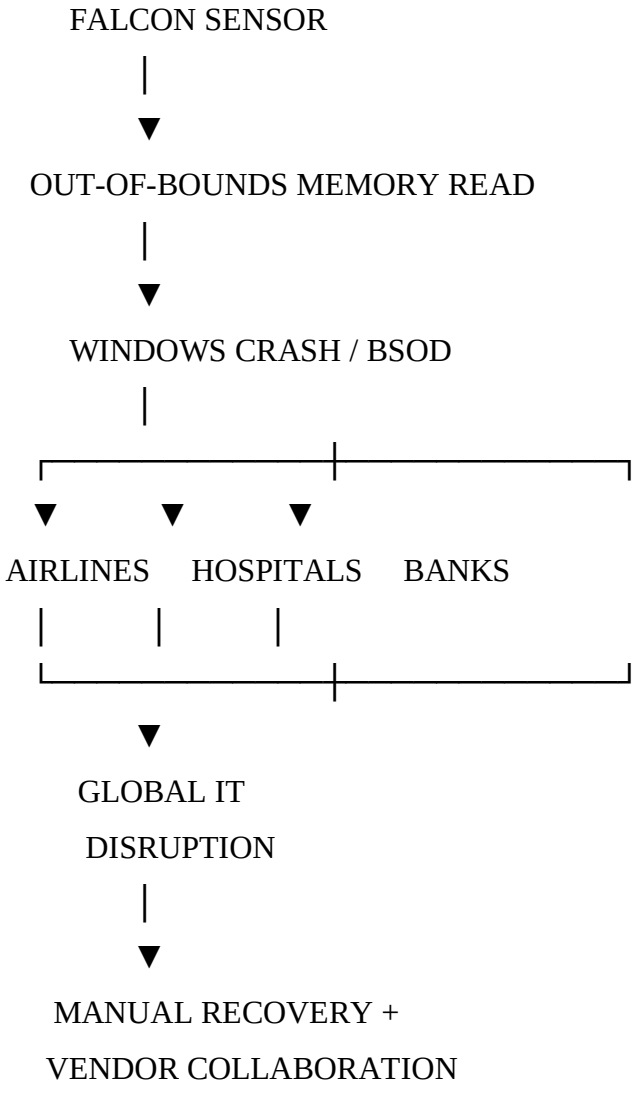
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▼

DEFECTIVE UPDATE

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▼



Root Cause Analysis

The underlying cause of the problem can be analyzed in terms of technology, procedure, and systems approach.

4.1 Technical Root Cause

The direct technical reason was due to bad data input format in the Rapid Response Content.

The Falcon sensor was engineered to work with a specific number of input variables.

Nevertheless, the troublesome information included another data field.

The Content Validator was supposed to recognize the discrepancy before the content made it into production systems. Due to the failure of validation, the faulty content made its way into distribution.

A memory access problem occurred when the Falcon sensor tried to analyze it.

It led to an exception that caused the Windows operating system to crash.

4.2 Validation Failure

This example highlights the significance of validation as one of the key controls within the software update chain.

The aim of validation is to make sure that the update meets the technical criteria of the system which is going to use it.

In addition, it is not that the update itself contained a flaw. Another important reason for the problem is the fact that the flaw managed to go beyond the security checks designed to avoid an incorrect update from reaching customers.

4.3 Deployment Risk

Another reason was the rapidity of deployment and its magnitude.

The timely nature of these updates is significant, as a delay in applying security patches leaves companies vulnerable to threats. This needs to be considered along with the probability of an error in the security update itself.

A better deployment approach could include:

Development ➜ Testing ➜ Internal release ➜ Canary release ➜ Limited customer release

➜ Monitoring ➜ Full deployment

CrowdStrike breach highlights the significance of such an approach.

4.4 Systemic Risk

This incident has also shown the danger associated with the dependency on technology.

Companies from various industries may have dissimilar business models; however, some use the same operational systems and cloud computing services and security solutions.

These include software supply chain risk, testing, and contingency planning, which were highlighted by the U.S. Government Accountability Office following the incident.

5. Impact Analysis

CrowdStrike outage hit several industries and showed the level of interconnection that exists in today's organizations.

5.1 Global IT Impact

The number of infected Windows computers was estimated at around 8.5 million. While this was a small percentage of the total number of Windows operating systems, those infected had high importance as CrowdStrike software was commonly used by

5.3 Healthcare

The healthcare industry was one of the most vulnerable industries during the outage.

The hospital administration, scheduling, patient records, communications, and many more tasks are becoming more reliant on computerized systems.

5.4 Financial Sector

Banks and financial institutions faced technology-related disturbances too.

Financial services rely upon uninterrupted access to their computer systems for customer support, business processing, communications, and transactions.

5.5 Government and Public Safety

Governments and public safety entities were also affected by the attacks. This was reported by the Congressional Research Service when they revealed that there were some disruptions to particular public safety systems.

Detection & Response

6.1 Detection

It was first apparent by the malfunctioning of the systems.

Windows failures were reported by organizations using the defective Falcon program. The number of failures occurring at one time suggested that this was not a case of a simple hardware or software failure.

CrowdStrike examined the problem and found the offending Rapid Response Content.

6.2 Immediate Response

CrowdStrike reversed the affected content around 05:27 UTC, which prevented the spread of the compromised content.

Although reversing the content could not instantly revive machines that had crashed.

Therefore, affected companies needed to carry out more recovery activities.

6.3 Microsoft Response

Microsoft collaborated with CrowdStrike and the affected firms for recovery purposes.

Microsoft also stated that they had mobilized hundreds of engineers and experts to assist their customers through the crisis. They also shared advice on how best to recover from the situation.

6.4 Recovery Challenges

A problem that emerged was the inability of certain affected systems to boot up properly.

This implies that there was the need for the administrators to use recovery environments to restore the computer systems.

This case showed how companies require a backup procedure to work in situations where the typical environment is not accessible.

6.5 Post-Incident Investigation

Once the crisis was past, CrowdStrike performed an in-depth analysis and shared their Root Cause Analysis.

The objectives of this analysis were to identify:

- What led to the poor content being made available?
- Reasons for non-detection of the problem through validation.
- How the issue spread to the customers' systems.
- Engineering controls that could be improved.
- Prevention of similar occurrences in future.

Security Measures & Mitigation

There are a few valuable security controls which can be implemented from the incident of CrowdStrike.

7.1 Stronger Content Validation

Security patches must be validated through multiple tests before deployment.

Validation should confirm:

- Correct input structure.
- Expected data types.
- Valid parameter ranges.
- Compatibility with supported systems.
- Safety precautions in unpredictable situations.

7.2 Staged Deployment

7.3 Fuzz Testing Critical updates should not immediately reach every endpoint.

Deployment architecture suggested by:

Development

↓

Automated Testing

↓

Security Testing

↓

Internal Environment

↓

Canary Group

↓

Limited Deployment

↓

Monitoring

↓

Large-Scale Deployment

In case there is any problem with the upgrade during the canary phase, then the deployment can be halted.

7.4 Fault Injection

Fault-injection testing deliberately introduces failures to determine how the system behaves**.**

Examples include:

- Invalid input. • Missing data.
- Unexpected parameters.
- Interrupted updates.
- Corrupted configuration.
- Resource exhaustion.

This helps determine whether the software fails safely.

7.5 Improved Error Handling

Security software should be designed so that an invalid configuration cannot cause an operating-system-wide crash.

The preferred behaviour should be:

Invalid Content → Reject Update → Continue Operating Safely rather than:

Invalid Content → System Crash → Organization Disrupted

7.6 Recovery and Rollback

Organizations should have mechanisms that allow problematic software updates to be rolled back quickly. Recovery tools should be accessible even when the affected endpoint cannot boot normally.

Lessons Learned

The CrowdStrike incident provides several important cybersecurity lessons.

8.1 Cybersecurity Products Can Become Critical Risks

Security software is normally considered a protective technology. However, because it operates deeply within operating systems, a defect can have extensive consequences.

Therefore, security software must be included in organizational risk assessments and business-continuity planning.

8.2 Availability Is Part of Cybersecurity

Cybersecurity is often described using the CIA triad:

- **Confidentiality**
- **Integrity**
- **Availability**

The CrowdStrike incident particularly demonstrates the importance of **availability**.

A system that is secure but unavailable cannot support normal business operations.

8.3 Software Supply-Chain Risk Is Important

Organizations depend on software produced and maintained by third parties.

A failure at a trusted vendor can therefore affect many downstream organizations.

Third-party software must be treated as part of the organization's security and operational risk environment.

8.4 Speed Must Be Balanced With Safety

Rapid security updates can protect organizations from emerging threats. However, speed without sufficient testing can introduce new risks.

Therefore, security vendors need to balance:

Rapid response + Strong validation + Controlled deployment

8.5 Disaster Recovery Must Be Practical

A disaster-recovery plan is useful only if it can actually be executed during a crisis.

Organizations should regularly test their recovery procedures and ensure that administrators have access to independent recovery mechanisms.

Recommendations

Based on the CrowdStrike incident, the following recommendations are proposed.

9.1 Recommendations for Security Vendors

1. **Implement multi-stage validation** before releasing security content.
2. **Use canary deployments** so that updates first reach a small controlled population.
3. **Conduct fuzz and fault-injection testing** to identify unexpected software behavior.

9.2Recommendations for Organizations

1. **Identify critical technology dependencies** and understand which vendors are essential to business operations.
2. **Maintain independent recovery mechanisms** for critical endpoints and servers.
3. **Regularly test business-continuity plans** instead of only documenting them.
4. **Conduct incident-response exercises** involving large-scale technology outages.

Conclusion

The CrowdStrike Global IT Outage of July 19, 2024, is an important real-world cybersecurity case because it demonstrates how a software defect can create global consequences without a malicious attacker.

The immediate technical problem occurred when defective Rapid Response Content was distributed through the CrowdStrike Falcon platform. A validation failure allowed the problematic content to reach production systems. The affected Falcon sensor processed an unexpected input structure, resulting in an out-of-bounds memory read and Windows system crashes.

The incident affected approximately 8.5 million Windows devices according to Microsoft, but its consequences were much broader because affected systems were used by organizations operating essential services.

The most important lesson is that **cybersecurity cannot focus only on preventing attacks**. It must also ensure that security systems themselves are reliable, resilient and capable of failing safely.

Disclaimer: This case study has been prepared solely for academic and educational purposes.

It is based on publicly available information concerning the CrowdStrike Falcon software update incident of July 19, 2024. The report does not represent CrowdStrike, Microsoft, any affected organization or any government agency. Certain technical concepts have been simplified for academic understanding and presentation. The analysis and recommendations are the author's academic interpretation of publicly available information.